

ECON 636 600: MACROECONOMIC THEORY I

Assignment 3 Solution

1. Recall that $F_1 = \alpha$ and F_j is increasing j and $\lim_{j \rightarrow \infty} F_j = \frac{\beta}{1-\alpha\beta} \equiv F_\infty$. Because $\log(\frac{A}{1+\beta F})$ and $\beta F \log(\frac{A\beta F}{1+\beta F})$ are continuous in F for $F \in [\alpha, F_\infty]$, we know that

$$\log\left(\frac{A}{1+\beta F_j}\right), \text{ and } \beta F_j \log\left(\frac{A\beta F_j}{1+\beta F_j}\right)$$

are bounded, i.e., there is $B > 0$ such that for all j ,

$$\begin{aligned} |\log\left(\frac{A}{1+\beta F_j}\right)| &< B, \\ |\beta F_j \log\left(\frac{A\beta F_j}{1+\beta F_j}\right)| &< B. \end{aligned}$$

To show the convergence of E_j , for any $\epsilon > 0$, pick a large $N > 0$ such that

$$\begin{aligned} & \left| \log\left(\frac{A}{1+\beta F_\infty}\right) + \beta \log\left(\frac{A}{1+\beta F_\infty}\right) + \beta^2 \log\left(\frac{A}{1+\beta F_\infty}\right) + \dots + \beta^N \log\left(\frac{A}{1+\beta F_\infty}\right) - \frac{\log\left(\frac{A}{1+\beta F_\infty}\right)}{1-\beta} \right| < \epsilon, \\ & \frac{\beta^{N+1} B}{1-\beta} < \epsilon, \end{aligned}$$

and

$$\begin{aligned} & \left| \log\left(\frac{A}{1+\beta F_j}\right) - \log\left(\frac{A}{1+\beta F_\infty}\right) \right| < \epsilon, \\ & \left| \beta F_j \log\left(\frac{A\beta F_j}{1+\beta F_j}\right) - \beta F_\infty \log\left(\frac{A\beta F_\infty}{1+\beta F_\infty}\right) \right| < \epsilon, \end{aligned}$$

for $j \geq N$. We have

$$\begin{aligned} & \left| E_j - \frac{\log\left(\frac{A}{1+\beta F_\infty}\right)}{1-\beta} - \frac{\beta F_\infty \log\left(\frac{A\beta F_\infty}{1+\beta F_\infty}\right)}{1-\beta} \right| \\ \leq & \left| \log\left(\frac{A}{1+\beta F_{j-1}}\right) + \beta \log\left(\frac{A}{1+\beta F_{j-2}}\right) + \beta^2 \log\left(\frac{A}{1+\beta F_{j-3}}\right) + \dots + \beta^{j-1} \log(A) - \frac{\log\left(\frac{A}{1+\beta F_\infty}\right)}{1-\beta} \right| \\ & + \left| \beta F_{j-1} \log\left(\frac{A\beta F_{j-1}}{1+\beta F_{j-1}}\right) + \beta(\beta F_{j-2} \log\left(\frac{A\beta F_{j-2}}{1+\beta F_{j-2}}\right)) + \dots + \beta^{j-2}(\beta F_1 \log\left(\frac{A\beta F_1}{1+\beta F_1}\right)) \right. \\ & \left. - \frac{\beta F_\infty \log\left(\frac{A\beta F_\infty}{1+\beta F_\infty}\right)}{1-\beta} \right|. \end{aligned}$$

I only show $\lim_{j \rightarrow \infty} |\log(\frac{A}{1+\beta F_{j-1}}) + \beta \log(\frac{A}{1+\beta F_{j-2}}) + \beta^2 \log(\frac{A}{1+\beta F_{j-3}}) + \dots + \beta^{j-1} \log(A) - \frac{\log(\frac{A}{1+\beta F_\infty})}{1-\beta}| = 0$, as the proof for the second term above is similar.

For $j \geq 2N$,

$$\begin{aligned}
& |\log(\frac{A}{1+\beta F_{j-1}}) + \beta \log(\frac{A}{1+\beta F_{j-2}}) + \beta^2 \log(\frac{A}{1+\beta F_{j-3}}) + \dots + \beta^{j-1} \log(A) - \frac{\log(\frac{A}{1+\beta F_\infty})}{1-\beta}| \\
\leq & |\log(\frac{A}{1+\beta F_{j-1}}) + \beta \log(\frac{A}{1+\beta F_{j-2}}) + \beta^2 \log(\frac{A}{1+\beta F_{j-3}}) + \dots + \beta^{j-1} \log(A) - \\
& \log(\frac{A}{1+\beta F_\infty}) - \beta \log(\frac{A}{1+\beta F_\infty}) - \beta^2 \log(\frac{A}{1+\beta F_\infty}) - \dots - \beta^N \log(\frac{A}{1+\beta F_\infty})| + \\
& |\log(\frac{A}{1+\beta F_\infty}) + \beta \log(\frac{A}{1+\beta F_\infty}) + \beta^2 \log(\frac{A}{1+\beta F_\infty}) + \dots + \beta^N \log(\frac{A}{1+\beta F_\infty}) - \frac{\log(\frac{A}{1+\beta F_\infty})}{1-\beta}| \\
\leq & |\log(\frac{A}{1+\beta F_{j-1}}) + \beta \log(\frac{A}{1+\beta F_{j-2}}) + \beta^2 \log(\frac{A}{1+\beta F_{j-3}}) + \dots + \beta^N \log(\frac{A}{1+\beta F_{j-N-1}}) - \\
& \log(\frac{A}{1+\beta F_\infty}) - \beta \log(\frac{A}{1+\beta F_\infty}) - \beta^2 \log(\frac{A}{1+\beta F_\infty}) - \dots - \beta^N \log(\frac{A}{1+\beta F_\infty})| + \\
& |\beta^{N+1} \log(\frac{A}{1+\beta F_{j-N-2}}) + \beta^{N+2} \log(\frac{A}{1+\beta F_{j-N-3}}) + \dots + \beta^{j-1} \log(A)| + \\
& |\log(\frac{A}{1+\beta F_\infty}) + \beta \log(\frac{A}{1+\beta F_\infty}) + \beta^2 \log(\frac{A}{1+\beta F_\infty}) + \dots + \beta^N \log(\frac{A}{1+\beta F_\infty}) - \frac{\log(\frac{A}{1+\beta F_\infty})}{1-\beta}| \\
\leq & |\epsilon + \beta\epsilon + \dots + \beta^N \epsilon| + |\beta^{N+1} B + \beta^{N+2} B + \dots + \beta^{j-1} B| + \epsilon \\
\leq & \frac{\epsilon}{1-\beta} + 2\epsilon
\end{aligned}$$

Since for $j \geq 2N$, $|\log(\frac{A}{1+\beta F_{j-1}}) + \beta \log(\frac{A}{1+\beta F_{j-2}}) + \beta^2 \log(\frac{A}{1+\beta F_{j-3}}) + \dots + \beta^{j-1} \log(A) - \frac{\log(\frac{A}{1+\beta F_\infty})}{1-\beta}| \leq \frac{\epsilon}{1-\beta} + 2\epsilon$, it satisfies the definition of convergence, i.e.,

$$\lim_{j \rightarrow \infty} \log(\frac{A}{1+\beta F_{j-1}}) + \beta \log(\frac{A}{1+\beta F_{j-2}}) + \beta^2 \log(\frac{A}{1+\beta F_{j-3}}) + \dots + \beta^{j-1} \log(A) = \frac{\log(\frac{A}{1+\beta F_\infty})}{1-\beta}$$

2. (50 points, 2012 Macro qualifier) Consider the following growth model where the planner chooses a sequence of consumption in order to

$$\max \sum_{t=0}^{\infty} \beta^t \ln(c_t),$$

$$s.t. \quad c_t + k_{t+1} \leq A_t(k_t)^\alpha, t \geq 0,$$

$$k_0 > 0.$$

Suppose that the sequence $\{A_t; t \geq 0\}$ follows a periodic pattern, that is, $A_t = 1$ for $t = 0, 2, 4, \dots$ and $A_t = \lambda > 1$ for $t = 1, 3, 5, \dots$

- (a) Set up the Bellman functional equations associated with odd period and even period.
- (b) Solve for the policy functions applicable to odd periods and policy functions applicable to even periods. (Conjecture and verify that the value function takes the form of $a_e + b_e \log(k)$ in even periods and $a_o + b_o \log(k)$ in odd periods, where a_e, b_e, a_o, b_o are parameters)
- (c) Show that the “steady state” of this dynamic model contains a two-period cycle.

(a)

$$V_e(k) = \max_{k'} \log(k^\alpha - k') + \beta V_o(k'),$$

$$V_o(k) = \max_{k'} \log(\lambda k^\alpha - k') + \beta V_e(k')$$

(b) Conjecture the functional forms

$$V_e(k) = a_e + b_e \log(k), \quad V_o(k) = a_o + b_o \log(k).$$

Verify the Bellman equation

$$a_e + b_e \log(k) = \max_{k'} \log(k^\alpha - k') + \beta(a_o + b_o \log(k')).$$

With optimal $k' = \frac{\beta b_o}{1 + \beta b_o} k^\alpha$, the right side of the Bellman equation is

$$a_e + b_e \log(k) = (\alpha + \alpha \beta b_o) \log(k) + \text{a constant}.$$

Hence $b_e = \alpha + \alpha\beta b_o$. Symmetrically verify the the Bellman equation

$$a_o + b_o \log(k) = \max_{k'} \log(\lambda k^\alpha - k') + \beta(a_e + b_e \log(k')),$$

we get $b_o = \alpha + \alpha\beta b_e$. We solve for b_e and b_o :

$$b_e = b_o = \frac{\alpha}{1 - \alpha\beta}.$$

Thus the policy function is

$$k' = \begin{cases} \alpha\beta k^\alpha, & t = \text{even}; \\ \alpha\beta \lambda k^\alpha, & t = \text{odd}, \end{cases}$$

we do not need to calculate parameters a_o or a_e to solve this problem.

- (c) If t is even, then $k_{t+2} = \alpha\beta\lambda(\alpha\beta k_t^\alpha)^\alpha = (\alpha\beta)^{1+\alpha}\lambda k_t^{\alpha^2}$, hence capital in even periods converge to

$$k_e^* = (\alpha\beta)^{\frac{1}{1-\alpha}} \lambda^{\frac{1}{1-\alpha^2}}.$$

Similarly,

$$k_o^* = (\alpha\beta)^{\frac{1}{1-\alpha}} \lambda^{\frac{\alpha}{1-\alpha^2}} < k_e^*.$$

Hence there is a cycle of capital stocks.

3. (20 points) Consider a growth model with foreign aid:

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq F(k_t, 1) + (1 - \delta)k_t + a, \\ & k_0 > 0. \end{aligned}$$

where $a > 0$ denotes the foreign aid. Let $k_{ss}(a)$ and $c_{ss}(a)$ be the capital stock and consumption in the steady state. Determine $\frac{dk_{ss}(a)}{da}$ and $\frac{dc_{ss}(a)}{da}$.

The steady state is characterized by

$$1 = \beta(F_k(k_{ss}, 1) + (1 - \delta))$$

$$c_{ss} = F(k_{ss}, 1) - \delta k_{ss} + a$$

Hence k_{ss} is unaffected by a , but $\frac{dc_{ss}(a)}{da} = 1$.